Sara Busatto, Ph.D.

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SUMMARY

- Tenure-track Assistant Professor, independent investigator, and head of the **Biological Nanoparticles Laboratory** within the Nanomedicine and Drug Targeting Unit at the Groningen Research Institute of Pharmacy.
- Independent and driven researcher with proven results in the isolation, characterization, loading, and engineering of biological cell-derived nanoparticles, in particular, extracellular vesicles.
- Experienced in **molecular biology and cellular biology** (e.g., qPCR analysis, miRNA isolation and characterization, cell transfection), **nanoparticle synthesis** (e.g., gold nanoparticles, liposomes), **loading** (e.g., small non-coding genetic material, standard chemotherapy) and **engineering** (e.g., targeting peptides, fusion proteins), **biophysical separation and concentration techniques** (e.g., tangential-flow filtration, ultracentrifugation, size-exclusion chromatography, density gradients and antibody-based immunoprecipitation), **multiple imaging modalities** (e.g., super-resolution confocal microscopy, atomic force microscopy and transmission microscopy), *in vivo* experiments and statistics.
- Solid and multidisciplinary knowledge in oncology; in particular, in **breast to brain metastasis** (e.g., premetastatic niche, early steps of the metastatic process, cancer cell migration), **cancer dormancy** (e.g., angiogenic switch, escape from cancer dormancy, breast cancer-associated mammary adipocytes) and simple and complex **endothelial barriers** (e.g., capillary endothelium, **blood-brain barrier**).
- Worked in four different countries and currently collaborates with international groups to study cell-derived nanoparticles known as extracellular vesicles (EVs), particularly their role in breast cancer brain metastasis. My research led to more than 45 publications, 2 patents, 1 funded postdoctoral research fellowship (NIH-T32), and over 20 presentations at national and international conferences.
- Works well in a team, has strong interpersonal skills, and communicates effectively with a diverse range of collaborators.

TECHNICAL SKILLS

- **Molecular Biology:** recombinant protein expression in mammal cells, molecular cloning, quantitative polymerase chain reaction (qPCR), PCR primer design, isolation of genomic DNA and mRNA, cDNA transcription, RNA interference/gene silencing using siRNA/shRNA, cell editing (plasmid transfection, synthetic miRNA mimic transfection, CRISPR/Cas9 technology).
- **Nanoparticle synthesis, loading and engineering:** synthesis of liposomes (sonication), cationic lipid-based nanoparticles (self-assembly), and gold nanoparticles (Turkevich method), biological and synthetic nanoparticles loading (passive incubation, electroporation, endogenous/cell-mediated loading, etc.) with small non-coding genetic material (miRNA, mRNA) and peptides (Cas9), engineering with fusion proteins and targeting peptides.
- **Flow cytometry**: proficient and independent user on Beckman Cytoflex and Thermo Fisher Scientific Attune NxT flow cytometers. In the process of acquiring and becoming a user of a nanoparticle flow cytometer, CytoFlex nanoflow from Beckman.
- **Biochemistry:** protein extraction and purification, Western blot, immunohistochemistry, co-immunoprecipitation, designing/validating immunoassays, affinity chromatography, ELISA, protein engineering (fusion proteins, targeting peptides).
- **Size and Density-based Separation and Concentration Methods:** tangential flow filtration (TFF), size-exclusion chromatography (SEC), bottom-up density gradients (both sucrose and iodixanol), ultracentrifugation (UC).
- **Biophysical Characterization Methods:** quartz crystal microbalance (QCM), dynamic Light Scattering (DLS), nanoparticle tracking analysis (NTA), surface plasmon resonance (SPR), transendothelial electrical resistance (TEER) and colorimetric nanoplasmonic.

- **Histology/imaging:** histology (e.g., H&E staining, Oil red O staining and Giemsa staining), immune histochemistry and immunofluorescence in various cells and tissues.
- **Microscopy:** laser confocal, fluorescent, atomic force (AFM), and bright field microscopy.
- **Cell biology:** Handling and maintaining different mammalian cells, isolation of primary cells (macrophages, primary brain microvascular endothelial cells, adipocytes, etc.). Transient and stable cell transfection, cell culture media optimization, electroporation, transcytosis assay (transwell), luciferase assay, cell migration/scratch assay, cell proliferation (BrdU, EdU, etc.) and cell viability assays.
- **Animal models:** animal handling techniques, anesthesia and analgesia administration, injection (intraperitoneal, intravenous, subcutaneous, intramuscular, retro-orbital), intracardiac injection, blood collection (intracardiac puncture) and organ and tissue isolation.

POSTDOCTORAL TRAINING AND EDUCATION

Harvard Medical School and Boston Children's Hospital, Boston, MA, United States
Postdoctoral Fellow in the Vascular Biology Program June 2020 – September 2024
Moses Laboratory
Vascular Biology Program
Boston Children's Hospital, Harvard Medical School
Boston, Massachusetts, United States
Advised by Prof. Marsha Moses

Mayo Clinic, Jacksonville, FL, United States
Postdoctoral Fellow April 2019 – June 2020
Nanomedicine and Extracellular Vesicles Laboratory
Mayo Clinic, Jacksonville; Florida, United States
Advised by Prof. Joy Wolfram

University of Brescia, Brescia, Italy
Ph.D. in Health Technology November 2015 – March 2019
- Visiting Ph.D. Student Fellowship Nanomedicine and Extracellular Vesicles Laboratory, Mayo Clinic, Jacksonville, Florida, United States
- Visiting Ph.D. Student Fellowship, University of Brescia, Brescia, Italy
- Ph.D. Fellowship, Italian Ministry of Education and University of Brescia, Brescia, Italy

Master of Science in Medical Biotechnology with honors October 2013 – May 2015
- Thesis Abroad Project, Italian Ministry of Education and University of Brescia, Brescia, Italy

University of Trieste, Trieste, Italy

Bachelor of Science in Biomedical Sciences with honors October 2008 – October 2011

PROFESSIONAL EXPERIENCE

University of Groningen, Groningen, The Netherlands
Assistant Professor and Independent Investigator. November 2024 – Present
- Coordinates independent research lines on:
  - (i) The role of cancer-derived extracellular vesicles (EVs) in the very early steps that lead to the formation of breast to brain metastasis. In particular, the preparation of a brain microenvironment, better known as the pre-metastatic niche, favors the survival and growth of metastatic cells.
  - (ii) Loading and engineering EV-mimetics as novel biocompatible actively targeted drug delivery vehicles.

(iii) Studying the formation and composition of the biomolecular corona of cancer-derived EVs.

- Contributes to collaborative projects on the role of EVs in Parkinson's disease and the use of EV-mimetics as vehicles for the delivery of protein-based therapeutics.
- Coordinates and executes research on developing novel and scalable engineering methods applicable to EVs and EV-mimetics in order to create drug delivery vehicles able to target primary and metastatic breast cancer cells.
- Proprietary method for the scalable and rapid synthesis of cell-membrane-derived EV-mimicking nanoparticles for drug delivery purposes.
- Oversees lab operations, performing duties of Lab Manager and Lab Safety Officer, as well as mentors and supervises the daily activities of one Ph.D. student, four MSc students, and one research assistant.

Boston Children's Hospital and Harvard Medical School, Boston, MA
Research Fellow in Vascular Biology | Marsha Moses, Ph.D. June 2020 – September 2024

- Coordinated and executed research on the very early steps that lead to the formation of breast to brain metastasis. In particular, the preparation of a brain microenvironment, better known as the pre-metastatic niche, favors the survival and growth of metastatic cells.
- Contributed to a collaborative project on the breast cancer microenvironment, in particular, on the role played by primary mammary tissue-derived adipocytes in the escape from tumor dormancy.
- Coordinated and executed research on developing novel and scalable engineering methods applicable to EVs and EV-mimetics to create drug delivery vehicles able to target primary and metastatic breast cancer cells.
- Oversaw lab operations, performing duties of Lab Manager and Lab Safety Officer, as well as mentoring and supervising the daily activities of one new lab member (postdoctoral fellow) and one research assistant.

Mayo Clinic, Jacksonville, FL
Research Fellow in Nanomedicine and Extracellular vesicles | Joy Wolfram, Ph.D. April 2019 – June 2020

- Coordinated and executed research centered on the interaction between cancer-derived EVs, circulating low-density lipoproteins and monocytes in the preparation of the brain pre-metastatic niche.
- Coordinated and executed research aimed at the development of a quick and simple method for the passive loading of proteins into EVs.
- The results from both these projects are reported in two first-author papers that were published in 2020 and 2021 in the *Journal of Nanobiotechnology* and in *Pharmaceuticals*, respectively.
- Collaborated in a research project showing that biogenic nanoparticles derived from patient lipoaspirate fluid mirror the activity of the EVs secreted from patient-matched adipose-derived stem cells representing a potential therapeutic strategy in regenerative nanomedicine.
- Oversaw lab operations performing duties of Lab Manager and Lab Safety Officer and mentored and supervised the daily activities of four new lab members (one graduate student and three postdocs).
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University of Brescia, Brescia, Italy
Graduate Student in Health Technology | Paolo Bergese, Ph.D. November 2015 – March 2019

- Spent two years as a Ph.D. student in Dr. Paolo Bergese's laboratory (University of Brescia, Brescia, Italy) and one year as a predoctoral student in Dr. Joy Wolfram's laboratory (Mayo Clinic, Jacksonville, Florida, USA).
- In Dr. Bergese's Laboratory. Coordinated and executed research on EV chemico-physical properties and engineering strategies. We developed a colorimetric nanoplasmonic assay to determine the purity and titration of unknown EV samples.

- In Dr. Wolfram's Laboratory. Developed a novel, scalable, and standardizable protocol for the purification of biological nanoparticles (e.g., EVs and lipoproteins) from various biological fluids that became a featured article in *Cells*, a peer-reviewed journal.

LEADERSHIP

University of Groningen, Groningen, The Netherlands. 2024 – Present

Leading member of the Local Organization Committee of the Annual Meeting of the Netherlands Society of Extracellular Vesicles sponsored by the Netherlands Society of Extracellular Vesicles to be hosted in Groningen, The Netherlands – **Organizer**.

Member of the International Organization Committee of the Inaugural Workshop on the extracellular vesicles biomolecular corona sponsored by the International Society of Extracellular Vesicles held in Brescia, Italy– **Organizer**.

Boston Children's Hospital, Boston, MA, USA

Postdoc Association (PDA) Mentoring Committee – Co-Chair 2020 – Present

- Planning and organizing events involving career mentoring for the postdoc community of the Boston Children's Hospital and the Longwood Area.

Boston Children's Hospital, Boston, MA, USA

BCH Green Lab Recycling Committee – Co-Chair 2020 – Present

- Planning and organizing meetings with BCH administrative personnel, supply chain, and companies to promote green and sustainable recycling/reusing initiatives throughout the Boston Children's Hospital community.

Mayo Clinic, Jacksonville, FL, USA

Leader Launch Program Zoom Cohort – Attendee 2019

- Program designed to create higher levels of confidence and effectiveness in leadership organized by Mayo Clinic Office of Entrepreneurship, Office of Research Diversity and Inclusion and Center for Biomedical Discovery Partnering with Capita 3.

KEY PUBLICATIONS (H-INDEX: 26)

- J. Sesen#, T. Martinez#, **S. Busatto#,** L. Poluben, H. Nassour, C. Stone, K. Ashok, M. A. Moses, E. R. Smith and A. Ghalali. Azin1 level is increased in medulloblastoma and correlates with c-myc activity and tumor phenotype. *Journal of experimental and clinical cancer research* 44, 56, 2025. **#co-first authors**. Https://doi.org/10.1186/s13046-025-03274-1
- **S. Busatto**, t. H. Song, H. J. Kim, C. Hallinan, M. N. Lombardo, A. O. Stemmer-Rachamimov, K. Lee, M. A. Moses. Breast cancer-derived extracellular vesicles modulate the cytoplasmic and cytoskeletal dynamics of blood-brain barrier endothelial cells. *Journal of extracellular vesicles* 14(1), 2025. Https://doi.org/10.1002/jev2.70038
- P. Guo, **S. Busatto**, J. Huang, G. Morad, M. A. Moses. A facile magnetic extrusion method for preparing endosome-derived vesicles for cancer drug delivery. *Advanced Functional Materials*, 2020; PMC Journal – In Press. doi: 10.1002/adfm.202008326. Cover of the Volume 31, Number 44, October 2021 https://doi.org/10.1002/adfm.202170328.
- **S. Busatto*,** Y. Yang, S. A. Walker, I. Davidovich, W. H. Lin, L. Lewis-Tuffin, P. Z. Anastasiadis, J. Sarkaria, Y. Talmon, G. Wurtz, J. Wolfram*. Brain metastases-derived extracellular vesicles induce binding and aggregation of low-density lipoprotein. Journal of Nanobiotechnology, 2020; 18(1):162. doi: 10.1186/s12951-020-00722-2. *Corresponding authors

- **S. Busatto*,** G. Villanilam, T. Ticer, W. L. Lin, D. W. Dickson, A. E. Thompson, S. Shapiro, P. Bergese* and J. Wolfram*. Tangential Flow Filtration for Highly Efficient Concentration of Extracellular Vesicles from Large Volumes of Fluid. *Cells*, 2018; 7(12):273. doi:10.3390/cells7120273. *Corresponding authors

Complete list of publications:
https://scholar.google.com/citations?hl=en&user=jJRawkgAAAAJ&view_op=list_works&sortby=pubdate